

1.1 RESEARCH METHODOLOGY

For this systematic literature review (SLR) study, the researcher adopts a qualitative research approach, which is well-suited for exploring the complex and context-dependent nature of big data analytics in the education sector of developing countries. This methodology allows for an in-depth examination of the various factors influencing the adoption and implementation of big data analytics in education within these unique settings. Qualitative methods facilitate a detailed examination of the diverse perspectives and experiences of stakeholders, such as educators, policymakers, and students, which is essential for understanding how big data analytics is perceived and implemented across various cultural and institutional contexts (Townsend & Saunders, 2024). This approach allows the researcher to explore the complexities of regulatory decision-making processes, revealing insights that quantitative methods may miss, such as the motivations, challenges, and values shaping the adoption of big data analytics in the education sector. (Coleman, 2022). Qualitative research offers flexibility, enabling the study to evolve as new themes and patterns emerge from the literature, resulting in a deeper and more thorough understanding of the topic. (Yadav, 2022). By focusing on depth rather than breadth, the qualitative approach offers the ideal framework to thoroughly capture the complexities and nuances of using big data analytics to enhance education in resource-limited settings.

1.2 DATA SOURCES

Numerous studies on big data and big data analytics, particularly their role in supporting regulatory decision-making, have been published in journal articles and peer-reviewed

conference papers. These studies are accessible through various academic databases. For this review, Scopus, Google Scholar, ACM Digital Library, Wiley Online Library, and MDPI were chosen as the primary sources for identifying relevant studies. These databases are highly regarded and among the largest, offering extensive collections of journals, conference proceedings, and books covering aquaponics and related fields. A summary of the selected databases is provided in Table 3.1.

Table 3.1 Description of the selected academic databases

Database	Description of the database	Source
ACM Digital Library,	The ACM Digital Library is a comprehensive database of computing and information technology literature, which is managed by the Association for Computing Machinery (ACM). It provides for includes peer-reviewed journals, conference proceedings, technical magazines, and books, that cover a broad spectrum of topics in computer science and related fields. The platform includes also the tools for citation tracking and access to cutting-edge research.	https://dl.acm.org/
Wiley Online Library,	Wiley Online Library is a vast digital platform that offers access to a wide range of peer-reviewed journals, books, and research articles across diverse disciplines, that include science, technology, medicine, humanities, and social sciences. It provides academic and professional research with high-quality, reliable content and user-friendly tools for discovery and citation.	https://onlinelibrary.wiley.com/
MDPI,	Multidisciplinary Digital Publishing Institute (MDPI) that provides an open-access publishing that offers a broad spectrum of peer-reviewed journals across disciplines, including science, technology, medicine, and social sciences. Known for its rapid publication process, MDPI supports free access to high-quality research that fosters global knowledge sharing and collaboration.	https://www.mdpi.com/
Google Scholar,	Google Scholar is an online platform that contains nearly 389 million full-text documents from diverse authors and academic fields. Established in November 2014, it offers a broad spectrum of academic materials, including peer-reviewed journal articles, books, theses, dissertations, abstracts, patents, legal reports, and conference papers.	https://scholar.google.com.
Scopus,	Scopus is described as the rich academic databases that provides peer reviewed academic journal articles, books and conference proceedings launched in 2004. Scopus spans a broad range of disciplines, including scientific,	https://www.scopus.com

	technical, medical, and social sciences, as well as arts and humanities, offering a global view of research output (Kolajo, Daramola & Adebisi, 2019). It offers access to approximately over 12 million pieces of science and medical. Over 3500 peer-reviewed academic journals are indexed in Scopus.	
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1.3 SEARCH STRINGS

The search was conducted across five databases: Scopus, Google Scholar, ACM Digital Library, Wiley Online Library, and MDPI, as these databases index most high-impact journals and conferences. To combine the search strings, the Boolean operator “OR” and “AND” was adopted and used in the search operations of five databases. Search strings that combined keywords that were based on the research questions were created to search and find relevant publications and the following keywords were used. The search terms were carefully crafted to encompass the core themes of the research, such as: "big data analytics", AND "education" AND "regulatory decision-making" AND "developing countries" while employing Boolean operators to refine the search and enhance the specificity of results. To ensure relevance and timeliness, the review primarily considered sources published between 2019 and 2024, reflecting the most current developments in the field. This systematic approach to data sourcing and search term selection, aligned with PRISMA guidelines, ensured that the review included a diverse array of literature, allowing for a more nuanced analysis of the topic while maintaining transparency, rigor, and a solid foundation for evidence-based conclusions (Clarke, Healy, Lynch & Featherstone, 2023; Hirose & Creswell, 2023; Townsend & Saunders, 2024). Table 3.2 presents the search strings used for the specific databases. These search strings were carefully designed to match the review objectives, ensuring thorough coverage of the relevant literature.

Table 3.2: Databases and search strings

Database	Search string
Google scholar	"big data analytics", AND "education" AND "regulatory decision-making" AND "developing countries"
ACM digital library	"big data analytics", AND "education" AND "regulatory decision-making" AND "developing countries"
MDPI	"big data analytics", AND "education" AND "regulatory decision-making" AND "developing countries"
Wiley online library	"big data analytics", AND "education" AND s"regulatory decision-making" AND "developing countries"
Scopus	"big data analytics", AND "education" AND "regulatory decision-making" AND "developing countries"

1.4 SELECTION CRITERIA

The selection criteria for this SLR were designed to ensure a relevant and high-quality body of literature was included in the analysis. The criteria were divided into inclusion and exclusion categories. Inclusion and exclusion criteria serve to define the scope, boundaries, relevance, and quality of a study, ensuring that only suitable and credible publications are included in the review. When these are applied during the selection process, they ensure that the selected studies are appropriate and meet the requirements for review. The role of inclusion criteria is to help researchers identify relevant primary studies across various databases, while the exclusion criteria process will eliminate studies that do not fit the review's focus. The inclusion and exclusion criteria applied in this study are presented in Table 3.3.

1.4.1 Inclusion criteria

- i. **Publication date:** Studies published between 2019 and 2024 were included to ensure the findings are current and relevant to the study's objectives.
- ii. **Language:** Only articles published in English were considered to maintain consistency in comprehension and analysis.

- iii. **Types of studies:** Peer-reviewed journal articles, conference papers, and reputable grey literature specifically focused on big data analytics within the education sector of developing countries were included.
- iv. **Focus on regulatory decision-making:** Studies that specifically focus on or offer insights into the use of big data analytics for regulatory decision-making in education were prioritised.
- v. **Empirical evidence:** Research that provided empirical data, case studies, or theoretical frameworks related to the application of big data in educational settings was included.

1.4.2 Exclusion criteria

- i. **Publication type:** Articles that were not peer-reviewed, including opinion pieces, editorials, and non-academic reports, were excluded to maintain the credibility and academic rigor of the sources.
- ii. **Irrelevant topics:** Studies that did not specifically focus on big data analytics or were centered on unrelated sectors or technologies were excluded to maintain the study's focus on the relevant field.
- iii. **Geographical focus:** Research that did not pertain to developing countries or did not provide context applicable to these regions was excluded.
- iv. **Outdated studies:** Articles published before 2018 were excluded to ensure the review captures the latest advancements and ongoing discussions in the field.
- v. **Insufficient data:** Studies that lacked sufficient empirical evidence or detailed analysis that were related to the research questions were excluded to maintain the quality of the review.

Table 3.3 Inclusion and Exclusion Criteria

Inclusion Criteria	Exclusion Criteria
Peer reviewed.	Papers not peer-reviewed, such as opinion pieces, editorials, or non-academic reports, were excluded to ensure the credibility of the sources.
Geographical focus on developing countries	Research from developed countries or did not provide context applicable to these regions was excluded.
Papers published between 2019 to 2024	To ensure the review reflects the most recent advancements and discussions in the field, articles published before 2019 were excluded. This approach aligns with the practice of prioritising recent studies to maintain the relevance and credibility of the research.
Papers that focus on big data analytics, technologies and more especially in education sector	Studies that did not specifically address big data analytics or were focused on unrelated sectors or technologies were excluded.
Papers written in English	Non English papers excluded

1.5 ARTICLE SCREENING AND CODING

The process of screening and coding articles played a crucial role in maintaining the relevance and quality of the literature included in this systematic literature review (SLR). To manage the retrieved articles efficiently, they were first compiled into a Microsoft Excel spreadsheet. The initial step involved removing duplicate entries to refine the dataset. Next, the researcher applied predefined inclusion and exclusion criteria by reviewing the titles and abstracts to assess their relevance. This step helped identify studies that directly addressed the research questions and contributed meaningful insights into big data analytics in education within developing countries.

For articles that passed this stage, full texts were obtained for a more in-depth evaluation. Each study was carefully examined to ensure its alignment with the review's objectives, particularly regarding methodological rigor and relevance to key research themes. Articles that met the inclusion criteria were systematically coded and assigned unique identifiers, such as Article 1 or Article 2, allowing for structured analysis and comparison. This systematic approach ensured a well-organised selection process, providing a solid foundation for synthesising findings in the review.

1.6 THE PAPER SELECTION PROCESS

In this study, a variety of data sources were utilised to ensure a comprehensive understanding of big data analytics in the education sector of developing countries, adhering to the guidelines set out by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA). Key academic databases, including the ACM Digital Library, Wiley Online Library, MDPI, Google Scholar, and Scopus, were primarily employed to identify a wide range of peer-reviewed articles, conference papers, and relevant grey literature that address the intersection of big data and education.

Each database was selected for its unique strengths, with ACM Digital Library focusing on computing and technology, providing access to a vast collection of full-text articles and bibliographic literature in these fields. Wiley Online Library offers multidisciplinary research across various academic disciplines, facilitating access to a broad spectrum of scholarly literature, MDPI provides open-access journals that cover innovative applications of data analytics, promoting the dissemination of research findings, Google Scholar hosts a vast array of scholarly literature, including peer-reviewed journal articles, books, theses, dissertations, abstracts, patents, court reports, and conference papers and finally Scopus which is a comprehensive abstract and citation database that covers a wide range of subjects, including science, technology, medicine, social sciences, and more. By leveraging these diverse sources, the study aimed to capture a holistic view of the current landscape of big data analytics in education within developing countries

The PRISMA flow chart in Figure 3.1 illustrates the systematic selection process for this review, detailing that a total of 62 full-text articles were assessed for eligibility, and this is not final inclusion number. After careful evaluation against the inclusion and exclusion criteria, 40 articles were ultimately incorporated into the systematic literature review. This structured approach ensured that the selected studies provided relevant and high-quality insights into big data analytics in the education sector of developing countries.

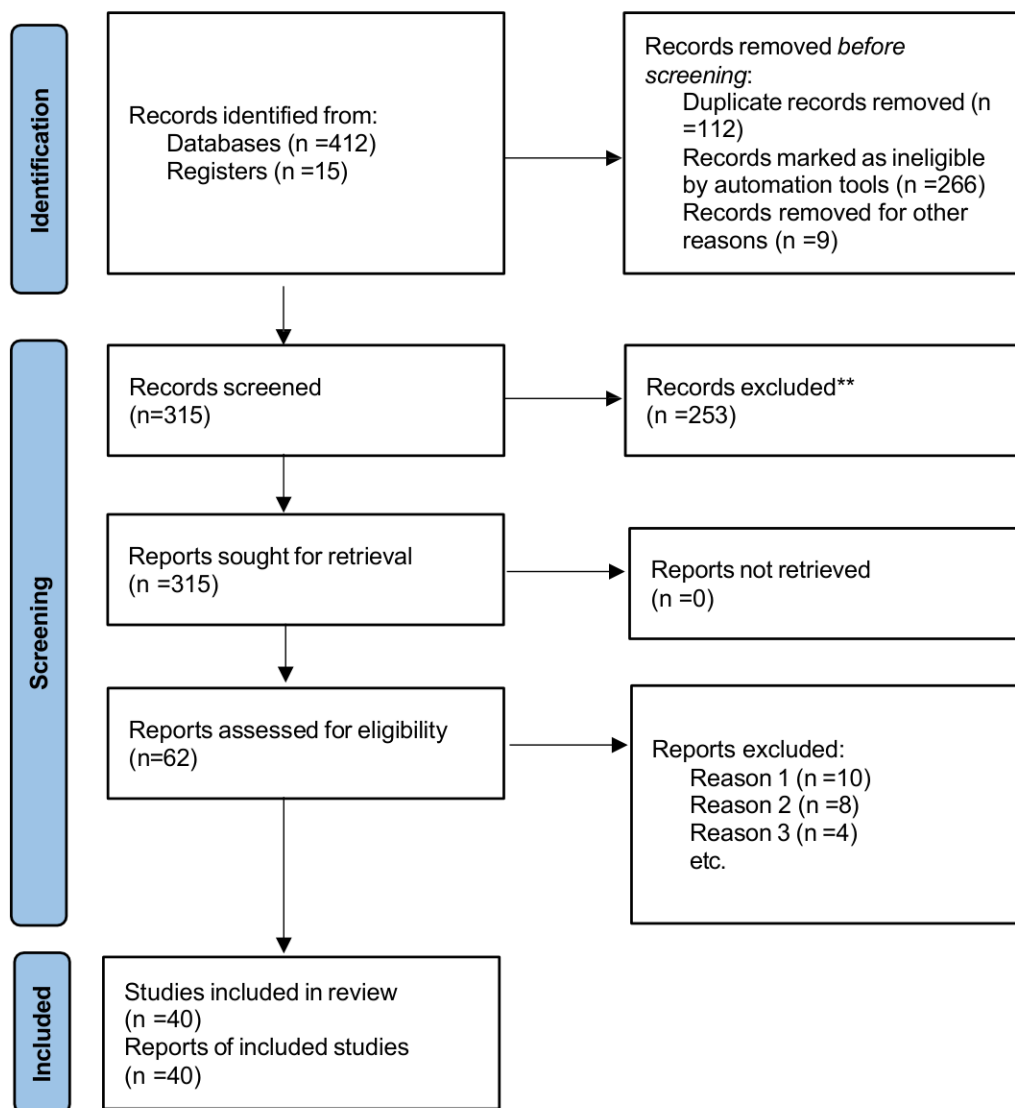


Figure 3.1: PRISMA flow chart for the publication selection process

1.7 RIGOR AND QUALITY ASSESSMENT OF CHOSEN ARTICLES

In this systematic literature review, each selected study underwent a rigorous evaluation to ensure the credibility and reliability of the findings. The assessment focused on several key aspects. Namely: (1) Methodological Rigor which evaluates the robustness of the research design and the appropriateness of the methods used, (2) Clarity of Research Questions which determines whether the studies clearly defined their research objectives, (3) appropriateness of Research Design assesses if the chosen design effectively addressed the research questions and (4) Relevance of Data Analysis Techniques that ensures that the data analysis methods were suitable for the study's aims.

To facilitate this evaluation, tools such as the Critical Appraisal Skills Programme (CASP) checklists were utilised. These checklists provide a structured framework for critically appraising both qualitative and quantitative studies, focusing on aspects like sample size, data collection methods, and transparency in reporting (CASP, 2018).

Additionally, the reliability and validity of the instruments used in the studies were examined, along with potential biases and conflicts of interest. This comprehensive quality assessment ensured that only high-quality articles were included in the review, contributing to a nuanced understanding of the strengths and limitations of the existing literature on big data analytics in the education sector (Gupta, Walia, Singh & Gupta, 2023). Ultimately, this approach to rigor and quality assessment strengthened the foundation of the review, allowing for more meaningful conclusions and recommendations to be drawn from the selected studies. Table 3.4 shows the quality assessment of publications using the CASP checklist.

Table 3.4: Quality assessment of publications using the CASP checklist

CASP Criteria	Assessment	Comments
Clear research questions	Yes/No	Questions were well-defined.
Appropriate methodology	Yes/No	Methodologies were suitable.
Rigorous sampling	Yes/No	Adequate sample sizes were used.
Data collection methods	Yes/No	Methods were well-documented.
Ethical considerations	Yes/No	Ethical approval noted.
Analysis of data	Yes/No	Analysis methods were clear.
Findings square with data	Yes/No	Findings were supported by data.
Limitations discussed	Yes/No	Limitations were highlighted
Contribution to knowledge	Yes/No	Valuable insights were provided
Clarity of reporting	Yes/No	Reporting was clear